

Robot Kinematics and Dynamics

Lecture 2A

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Some slides sourced from Lynch and Park, Modern Robotics
(credited on slide)

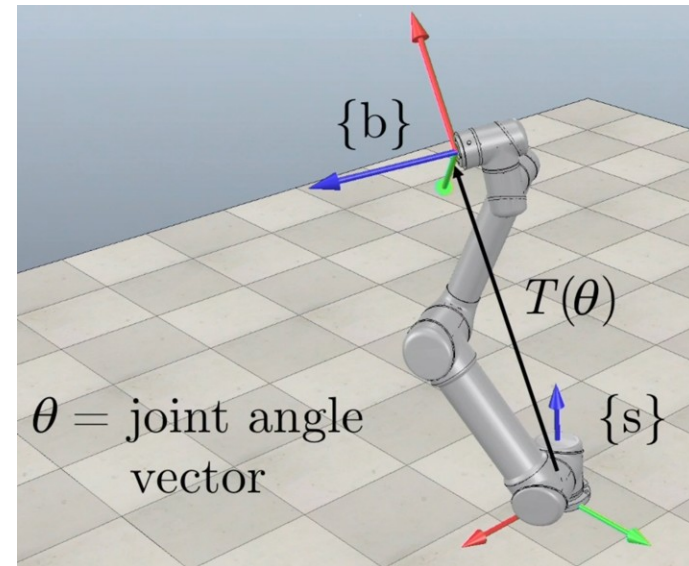
Important concepts, symbols, and equations

Forward kinematics of a serial chain:

Given

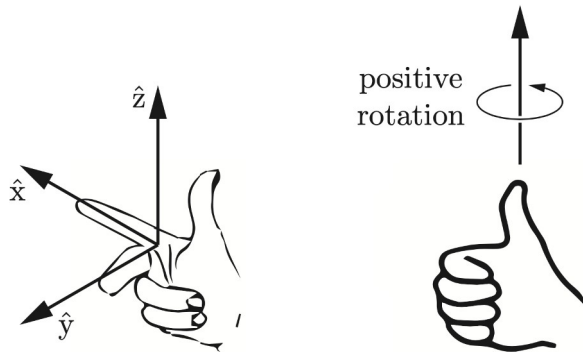
- $M = T_{sb}(0) \in SE(3)$, the configuration of the end-effector frame $\{b\}$ at the home configuration $\theta = 0$,
- the screw axes for each joint at $\theta = 0$, and
- the joint vector θ ,

find $T_{sb}(\theta) \in SE(3)$.



Important concepts, symbols, and equations

- We often define a fixed **space frame** $\{s\}$ and a **body frame** $\{b\}$ attached to some body of interest. All frames are *instantaneously stationary*.
- Right-handed frames, and right-hand rule for positive rotation.



Important concepts, symbols, and equations (cont.)

- For screw axes S expressed in $\{s\}$, the **product of exponentials** (PoE) in the space frame is

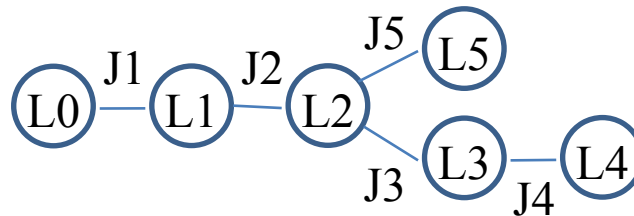
$$T(\theta) = e^{[S_1]\theta_1} \dots e^{[S_n]\theta_n} M$$

- For screw axes $\mathcal{B}$ expressed in $\{b\}$, the PoE is

$$T(\theta) = M e^{[\mathcal{B}_1]\theta_1} \dots e^{[\mathcal{B}_n]\theta_n}$$

Important concepts, symbols, and equations (cont.)

The **Universal Robot Description Format (URDF)** is an XML file describing the kinematics, inertial properties, and geometry of a tree-structured robot.



the **joint** element (kinematics)

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<joint name="joint2" type="revolute">
  <parent link="link1"/>
  <child link="link2"/>
  <origin rpy="0 1.5708 0" xyz="0 0.5 0">
  <axis xyz="0 1 0">
  <limit lower="-3.0" upper="3.0">
</joint>
```

the **link** element (inertial properties)

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<link name="link2">
  <inertial>
    <origin rpy="0 0 0" xyz="0 0 0.5">
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    <inertia ixx="1.0" ixy="0" ixz="0"
      iyy="2.0" iyz="0" izz="3.0"/>
    </inertial>
  </link>
```

Definition 1 *Euclidean transformations (also known as isometries) are transformations that preserve distances between pairs of points.*

$$||\psi(\mathbf{a}) - \psi(\mathbf{b})|| = ||\mathbf{a} - \mathbf{b}|| \quad (3.1)$$

Translations, $\psi(\mathbf{a}) = \mathbf{a} + \mathbf{t}$, are isometries, since

$$||\psi(\mathbf{a}) - \psi(\mathbf{b})|| = ||\mathbf{t} + \mathbf{a} - (\mathbf{t} + \mathbf{b})|| = ||\mathbf{a} - \mathbf{b}|| \quad (3.2)$$

Rigid body motions (Euclidean transformations / isometries)

- **Theorem:** Any rigid body motion can be expressed as an orthogonal transformation followed by a translation.

$$\psi(\mathbf{a}) = \mathbf{A}\mathbf{a} + \mathbf{t}$$

Orthogonal Matrices in 2D

In 2D, these constraints put together force $\mathbf{A}$ to be one of two types of matrices.

$$\underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_{\text{rotation, det}=+1} \text{ or } \underbrace{\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}}_{\text{reflection, det}=-1}$$

Under a rotation by angle θ ,

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

The reflection matrix above corresponds to reflection around the line with angle $\frac{\theta}{2}$ (verify). Note that two rotations one after the other give another rotation, while two reflections give us a rotation.

Orthogonal Matrices in 3D

Let us now construct some examples in 3D. Just as in 2D, rotations are characterized by orthogonal matrices with $\det = +1$. For orthogonal matrices, each column vector has length 1, and the dot product of any two different columns is 0. This gives rise to six constraints (3 pairwise dot product constraints, and 3 length constraints), so for a 3 dimensional rotation matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (3.14)$$

with 9 total parameters, there are really only three free parameters. There

- Rotation about z-axis by θ :

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Rotation about x-axis by θ :

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

Parameterizing Rotations in 3D

Recall that rotation matrices have the property that each column vector has length 1 and the dot product of any 2 different columns is 0. These 6 constraints leave only 3 degrees of freedom. Here are some alternative notations used to represent orthogonal matrices in 3-D:

- Euler angles which specify rotations about 3 axes
- Axis plus amount of rotation
- Quaternions which generalize complex numbers from 2-D to 3-D. (Note, a complex number can represent a rotation in 2-D)

We will use the axis and rotation as the preferred representation of an orthogonal matrix: $\mathbf{s}, \theta$, where $\mathbf{s}$ is the unit vector of the axis of rotation and θ is the amount of rotation.

Skew symmetric matrices can be used to represent “cross” products or vector products. Recall:

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \wedge \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

We define $\hat{\mathbf{a}}$ as:

$$\hat{\mathbf{a}} \stackrel{def}{=} \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

Thus, multiplying $\hat{\mathbf{a}}$ by any vector gives:

$$\begin{aligned} \hat{\mathbf{a}} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} &= \begin{bmatrix} -a_3 b_2 + a_2 b_3 \\ a_3 b_1 - a_1 b_3 \\ -a_2 b_1 + a_1 b_2 \end{bmatrix} \\ &= \mathbf{a} \wedge \mathbf{b} \end{aligned}$$

Consider now, the equation of motion of a point q on a rotating body:

$$\dot{\mathbf{q}}(t) = \boldsymbol{\omega} \wedge \mathbf{q}(t)$$

where the direction of $\boldsymbol{\omega}$ specifies the axis of rotation and $\|\boldsymbol{\omega}\|$ specifies the angular speed. Rewriting with $\hat{\boldsymbol{\omega}}$

$$\dot{\mathbf{q}}(t) = \hat{\boldsymbol{\omega}} \mathbf{q}(t)$$

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$$\dot{\mathbf{q}}(t) = \hat{\boldsymbol{\omega}} \mathbf{q}(t)$$

The solution to this differential equation uses the matrix exponential

$$\mathbf{q}(t) = e^{\hat{\boldsymbol{\omega}} t} \mathbf{q}(0)$$

Where,

$$e^{\hat{\boldsymbol{\omega}} t} = \mathbf{I} + \hat{\boldsymbol{\omega}} t + \frac{(\hat{\boldsymbol{\omega}} t)^2}{2!} + \frac{(\hat{\boldsymbol{\omega}} t)^3}{3!} + \dots$$

Collecting the odd and even terms in the above equation, we get to **Roderigues Formula** for a rotation matrix **R**.

$$\begin{aligned}\mathbf{R} &= e^{\phi \hat{\mathbf{s}}} \\ &= \mathbf{I} + \sin \phi \, \hat{\mathbf{s}} + (1 - \cos \phi) \hat{\mathbf{s}}^2\end{aligned}$$

Here **s** is a unit vector along ω and $\phi = \|\omega\|t$ is the total amount of rotation. Given an axis of rotation, **s**, and amount of rotation ϕ we can construct $\hat{\mathbf{s}}$ and plug it in.

The composition of two isometries is an isometry

$$\psi_1(\mathbf{a}) = \mathbf{A}_1\mathbf{a} + \mathbf{t}_1 \qquad \psi_2(\mathbf{a}) = \mathbf{A}_2\mathbf{a} + \mathbf{t}_2.$$

$$\begin{aligned}\psi_1 \circ \psi_2(\mathbf{a}) &= \mathbf{A}_1(\mathbf{A}_2\mathbf{a} + \mathbf{t}_2) + \mathbf{t}_1 \\ &= \mathbf{A}_1\mathbf{A}_2\mathbf{a} + \mathbf{A}_1\mathbf{t}_2 + \mathbf{t}_1 \\ &= (\mathbf{A}_1\mathbf{A}_2)\mathbf{a} + (\mathbf{A}_1\mathbf{t}_2 + \mathbf{t}_1) \\ &= \mathbf{A}_3\mathbf{a} + \mathbf{t}_3\end{aligned}$$

where $\mathbf{A}_3 = \mathbf{A}_1\mathbf{A}_2$ and $\mathbf{t}_3 = \mathbf{A}_1\mathbf{t}_2 + \mathbf{t}_1$.

Homogeneous coordinates

- Instead of using n coordinates for n -dimensional space, we use $n+1$ coordinates.

Key rule

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Important concepts, symbols, and equations

- The **special Euclidean group** $SE(3)$ is a matrix Lie group also known as the group of rigid-body motions or **homogeneous transformation matrices** in $\mathbb{R}^3$.

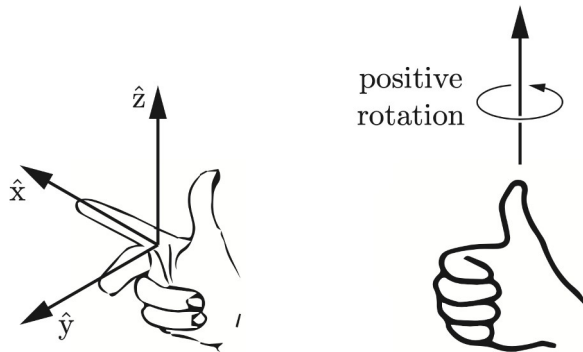
$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ r_{31} & r_{32} & r_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3) \quad R \in SO(3) \text{ and } p \in \mathbb{R}^3$$

- The inverse of $T \in SE(3)$ is

$$T^{-1} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations

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Important concepts, symbols, and equations (cont.)

- Three uses of HT matrices:

1. Represent a configuration. T_{ab} represents frame {b} relative to {a}.
2. Change the reference frame of a vector or frame.

$$T_{ab}T_{bc} = T_{a\cancel{b}}T_{\cancel{b}c} = T_{ac}$$
$$T_{ab}v_b = T_{a\cancel{b}}v_{\cancel{b}} = v_a$$

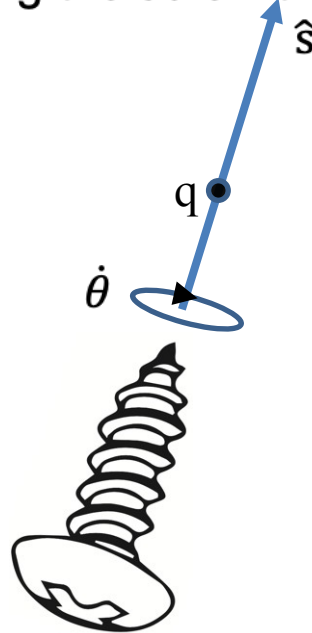
v should be written in
homogeneous coordinates,
 $v = [v_1 \ v_2 \ v_3 \ 1]^T$.

3. Displace a vector or frame. $T = (R, p) = \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta)$

$$\text{Trans}(p) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{Rot}(\hat{\omega}, \theta) = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations (cont.)

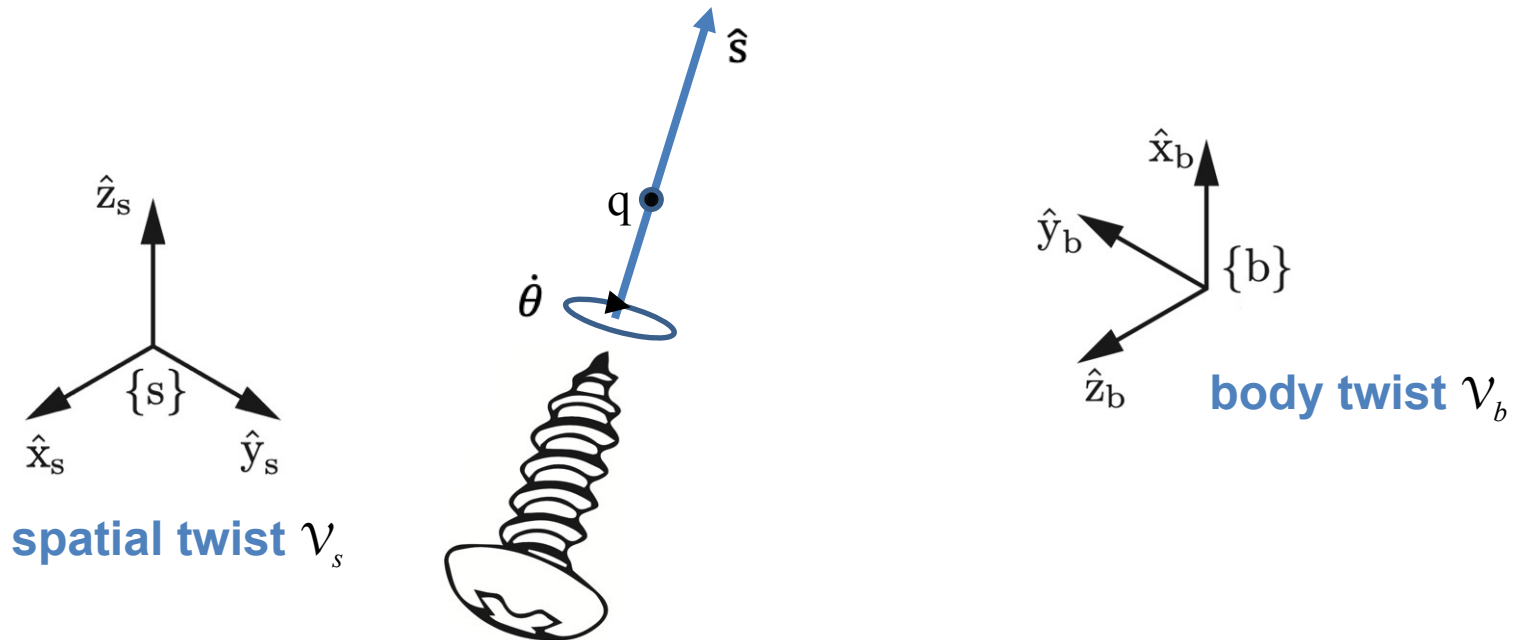
Any rigid-body velocity can be represented as a **screw axis** (a direction $\hat{s} \in S^2$, a point $q \in \mathbb{R}^3$ on the screw, and the **pitch** (linear speed/angular speed) of the screw h), plus the speed along the screw $\dot{\theta}$.



If h is infinite, $\dot{\theta}$ is the linear speed. Otherwise, it is the angular speed.

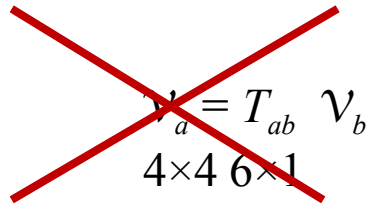
Important concepts, symbols, and equations (cont.)

The **twist** $\mathcal{V}_a = (\omega_a, v_a) \in \mathbb{R}^6$ is the angular velocity expressed in $\{a\}$ and the linear velocity of the origin of $\{a\}$ expressed in $\{a\}$.



Important concepts, symbols, and equations (cont.)

- To transform a twist from one frame to another,


$$\mathcal{V}_a = T_{ab} \mathcal{V}_b$$

$4 \times 4 \quad 6 \times 1$

$\mathcal{V}_a = [\text{Ad}_{T_{ab}}] \mathcal{V}_b$, where the **adjoint representation** of $T = (R, p)$ is

$$[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

Important concepts, symbols, and equations (cont.)

Matrix representation of twists:

$$T_{sb}^{-1} \dot{T}_{sb} = [\mathcal{V}_b] = \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix} \in se(3)$$

$$\dot{T}_{sb} T_{sb}^{-1} = [\mathcal{V}_s] = \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix} \in se(3)$$

where $se(3)$ is the Lie algebra of $SE(3)$ (the set of all possible $\dot{T}$ when $T = I$).

Important concepts, symbols, and equations (cont.)

Screws and twists:

- for a screw axis $\{q, \hat{s}, h\}$ with finite h ,

$$\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} = \begin{bmatrix} \hat{s} \\ -\hat{s} \times q + h\hat{s} \end{bmatrix}$$

- “unit” screw axis is $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$,
where either (i) $\|\omega\| = 1$ or
(ii) $\omega = 0$ and $\|v\| = 1$

- $\mathcal{V} = \mathcal{S}\dot{\theta}$

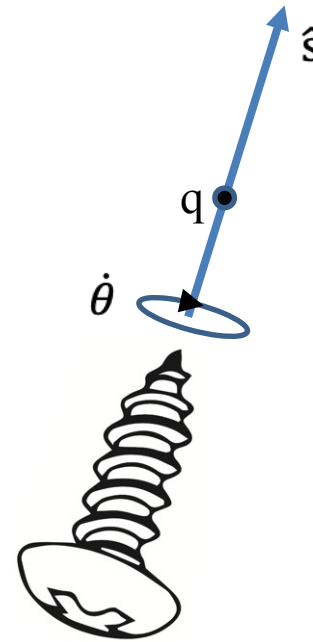
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- $\mathcal{V} = \mathcal{S}\dot{\theta}$



What is the dimension of the space of screws?

Important concepts, symbols, and equations

- A configuration can be represented by **exponential coordinates** $S\theta \in \mathbb{R}^6$: a screw axis S multiplied by the distance θ it is followed. (Equivalently, $\forall t$: a twist $\mathcal{V}$ and a time t it is followed.)
- As with rotations, we can define a matrix exponential and its inverse, the matrix log. The exponential “integrates a twist” for time 1, and the log finds the constant twist needed to achieve the displacement in time 1.

$$\begin{aligned} \exp : [S]\theta \in se(3) &\rightarrow T \in SE(3) \\ \log : T \in SE(3) &\rightarrow [S]\theta \in se(3) \quad \theta \in [0, \pi] \end{aligned}$$

Important concepts, symbols, and equations

For $S = (\omega, v)$, either

- $\|\omega\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega]\theta} & (I\theta + (1 - \cos \theta)[\omega] + (\theta - \sin \theta)[\omega]^2) v \\ 0 & 1 \end{bmatrix}$$

- or $\omega = 0$ and $\|v\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations (cont.)

- A **wrench** is $\mathcal{F} = (m, f) \in \mathbb{R}^6$. A linear force $f \in \mathbb{R}^3$ at r creates a moment $m = r \times f$.
- The dot product of a wrench and a twist is power: $P = \mathcal{V}^T \mathcal{F}$.
- The same wrench can be expressed in $\{a\}$ and $\{b\}$ as $\mathcal{F}_a$ and $\mathcal{F}_b$.
- Changing the frame of representation (power better be independent of the frame we use to represent twists and wrenches!):

$$\mathcal{V}_b^T \mathcal{F}_b = \mathcal{V}_a^T \mathcal{F}_a$$

$$\begin{aligned}\mathcal{V}_b^T \mathcal{F}_b &= ([\text{Ad}_{T_{ab}}] \mathcal{V}_b)^T \mathcal{F}_a \\ &= \mathcal{V}_b^T [\text{Ad}_{T_{ab}}]^T \mathcal{F}_a.\end{aligned}$$

$$\mathcal{F}_b = [\text{Ad}_{T_{ab}}]^T \mathcal{F}_a$$

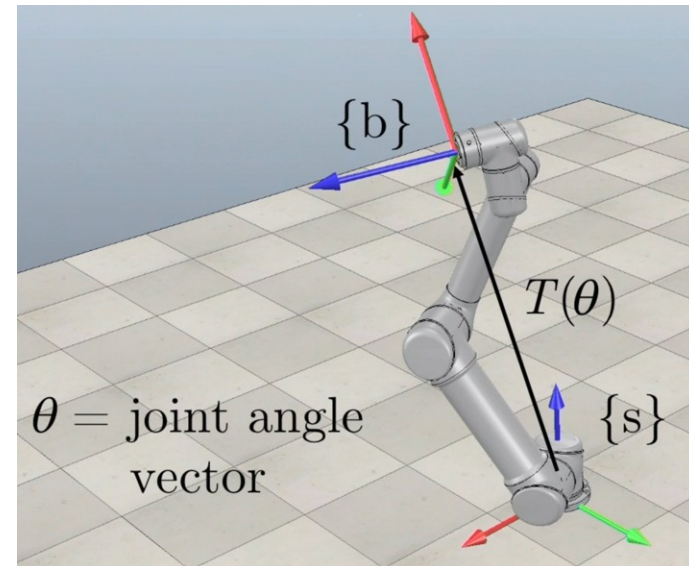
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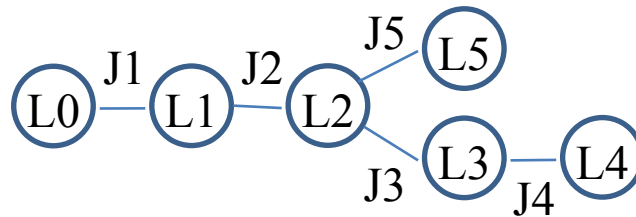
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